IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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April 14, 2026

Instructions and guidelines (read carefully)

Instructions

1. Insert your name as it appears on your Online Campus learner profile in the file name. Save the file as: **First Name Last Name M10U4 Capstone Project – e.g., Lilly Smith M10U4 Capstone Project.**
2. Write all your answers and upload your screenshots in this PowerPoint presentation in the slides provided.
3. Convert your final presentation into **.pdf** format and submit this **PDF document**. No other file types will be accepted.

Note: Please ensure that you have checked your course calendar for the due date for this assignment.

Guidelines

Note: If you do not have a desktop version, follow the instructions from the previous lab to access PowerPoint online.

1. Complete the slides with relevant details (you are free to add additional slides, charts, and tables).
2. Add any additional unused charts or tables to the appendix that you think are relevant to your analysis.

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Outline

- Executive summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

GitHub Link: https://github.com/Kleiner7/SpaceXCapstone/tree/main

Executive summary

- **Objective**: This project aims to predict the success of Falcon 9 first-stage landings, a key factor in reducing SpaceX launch costs through rocket reusability. The analysis uses data from the SpaceX API, supplemented with Falcon 9 launch information from Wikipedia.

- **Exploratory Data Analysis:** Static and interactive visualizations were used to identify patterns between landing success and variables such as launch site, payload mass, orbit type, and flight number. Dashboards built with Folium and Plotly Dash provided geographic insights into landing locations.

- **Machine Learning:** Models including logistic regression, support vector machines (SVM), decision trees, and K-nearest neighbors (KNN) were evaluated. Given similar performance across models, logistic regression was selected for its simplicity, efficiency, and interpretability.

- **Results:** Landing success can be predicted with reliable accuracy, supporting cost estimation and pricing decisions in the commercial launch market.

Introduction

- **Background:** SpaceX's Falcon 9 reduces launch costs by reusing its first-stage booster. Successful landings allow the rocket to be refurbished and flown again, lowering costs compared to traditional single-use rockets

- **Problem Statement:** Can we predict whether the Falcon 9 first stage will land successfully and develop a model to accurately estimate landing outcomes to support cost estimation and planning decisions

Section 1

Methodology

Methodology

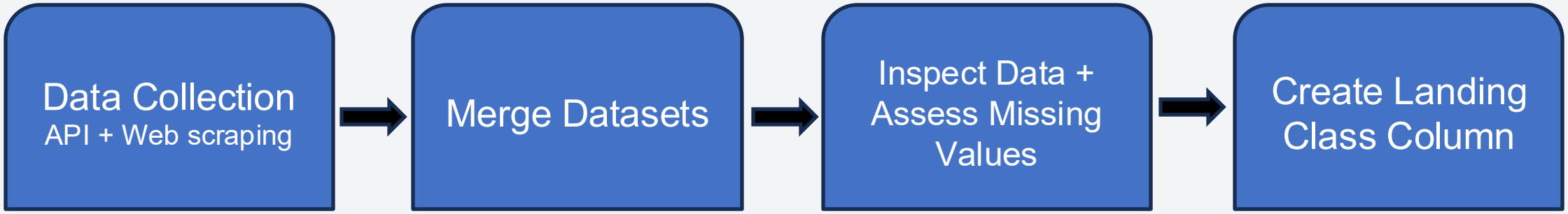
Executive summary

- **Data Collection:** SpaceX API data supplemented with Wikipedia launch data
- **Data Wrangling:** Cleaned and prepared data for analysis
- **EDA:** Used SQL and visualizations to identify key patterns
- **Visualization:** Built interactive maps and dashboards with Folium and Plotly Dash
- **Modeling:** Trained classification models (logistic regression, SVM, decision tree, KNN)
- **Evaluation:** Tuned and evaluated models to select the best performer

Data collection

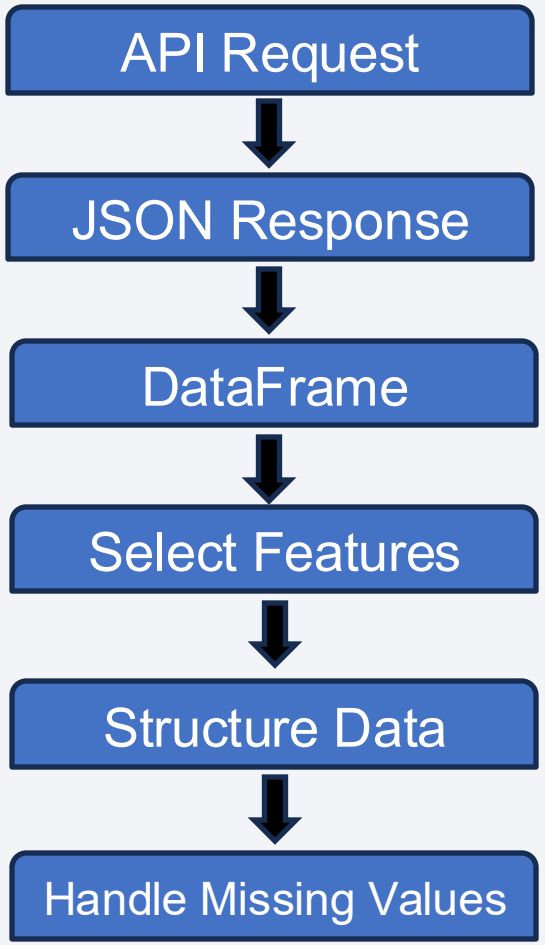
Data Collection Overview:

- **Sources:** SpaceX API and Wikipedia (Falcon 9 launch data)
- **Approach:** Collected structured data via API and supplemented with web-scraped data
- **Processing:** Merged datasets, assessed missing values, filled missing payload mass with mean values, and created a **Landing Class** variable (1 = success, 0 = failure)
- **Process Flow:**



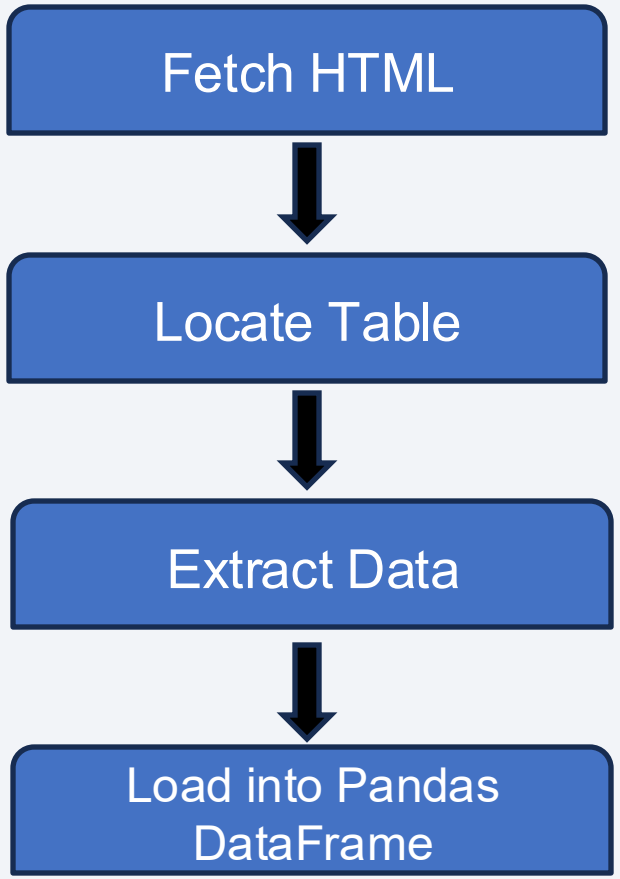
Data collection – SpaceX API

- **Source**: SpaceX API
- Sent API requests to retrieve launch data in JSON format
- Transformed the response into a Pandas DataFrame
- Selected and organized relevant features (Launch Site, Orbit, Booster Version, etc.) into a structured dataset
- Assessed and handled missing values during preprocessing
- External Reference: GitHub Repository – [SpaceX API Data Collection Notebook]



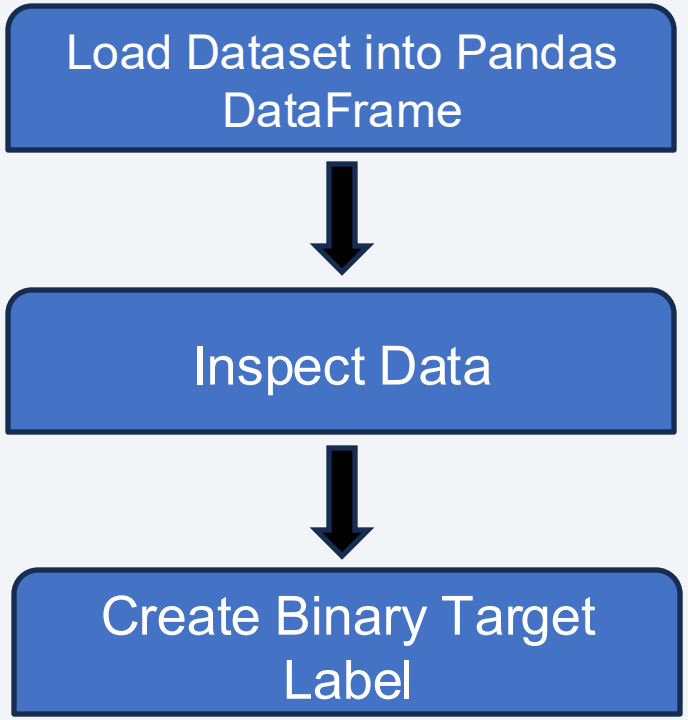
Data collection – Scraping

- **Source**: Wikipedia – List of Falcon 9 and Falcon Heavy Launches
- Extracted Falcon 9 launch data from publicly available Wikipedia tables
- Identified and selected relevant tables containing launch details
- Parsed and extracted structured data from HTML tables
- Transformed the extracted data into a Pandas DataFrame
- External Reference: GitHub Repository – SpaceX Webscraping Notebook



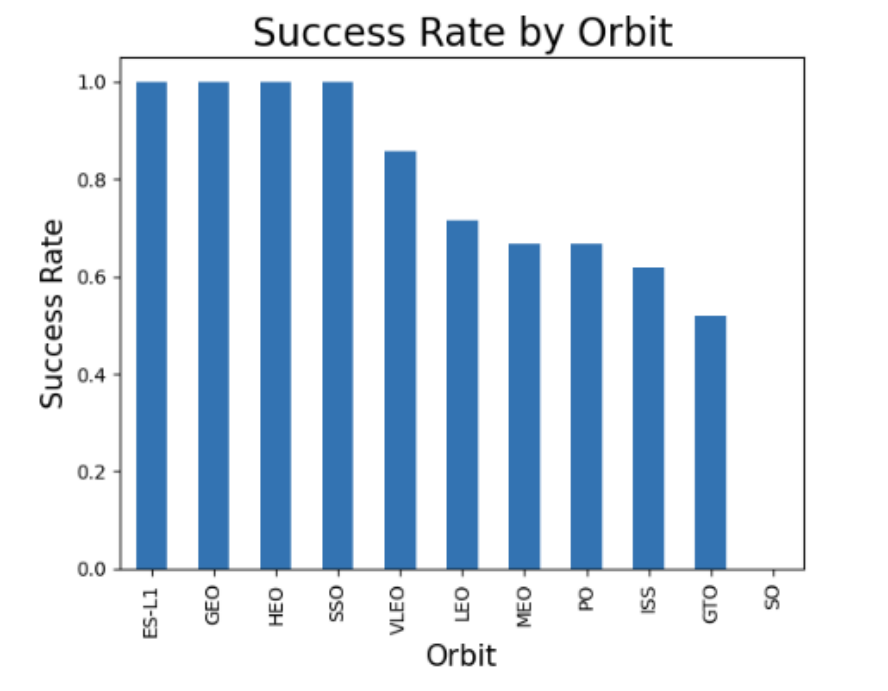
Data wrangling

- Loaded the dataset (from API and Wikipedia) into a Pandas DataFrame
- Assessed missing values and data structure
- Analyzed launch outcome categories
- Converted outcomes into a binary target label (1 = success, 0 = failure)
- Prepared the dataset for modeling
- External Reference: GitHub Repository – [SpaceX Data Wrangling Notebook]



EDA with data visualization

- Loaded the prepared dataset into a Pandas DataFrame
- Visualized key patterns using scatter plots, bar charts, and line charts
- Used scatter plots to analyze relationships between variables, bar charts to compare success rates across categories, and line charts to observe trends over time
- Analyzed how factors such as payload mass, orbit type, and launch site impact outcomes
- Identified trends and insights to support feature selection for modeling
- External Reference: GitHub Repository – [SpaceX EDA with Data Visualization Notebook]



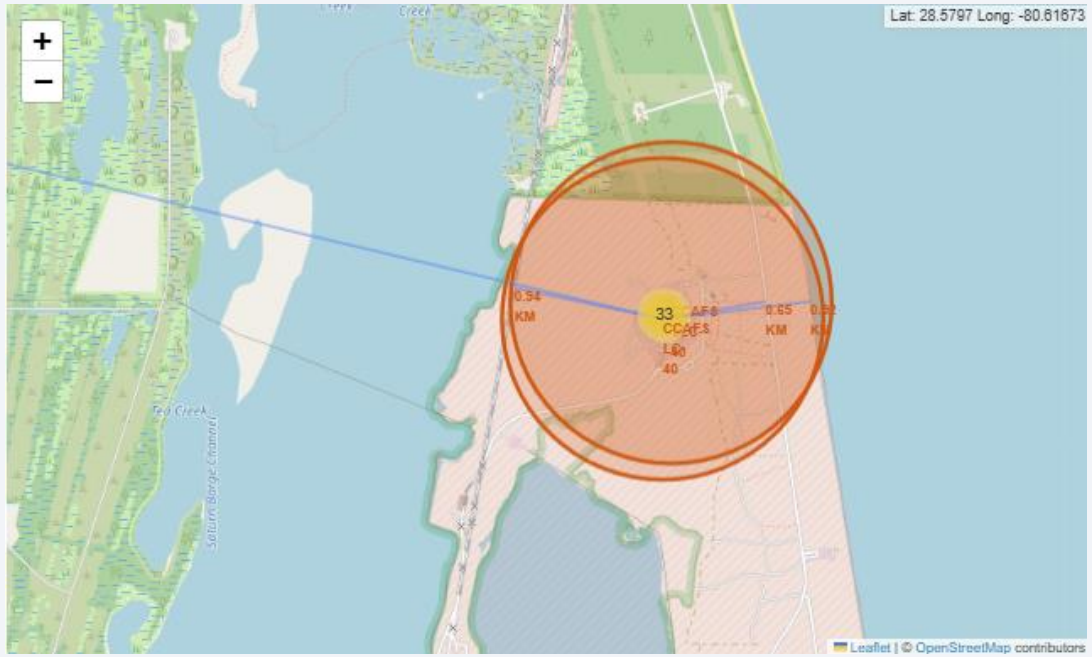
Orbits such as ES-L1, GEO, HEO, and SSO show the highest landing success rates

EDA with SQL

- Retrieved and filtered launch records by site, payload, and booster characteristics
- Calculated total and average payload mass for specific boosters and customers (e.g. NASA (CRS))
- Identified date of first successful drone ship landing and names of boosters which had success in ground pad landings within specific payload ranges (4000-6000 kg)
- Explored time-based trends, including successful landing outcomes with month and booster version in the year 2017
- Ranked count of landing outcomes within a selected date range
- External Reference: GitHub Repository – SpaceX EDA with SQL

Build an interactive map with Folium

- Used large circle markers to represent the location of each launch site and provide a clear geographic overview of all launch locations
- Used color-coded markers within each site (green = successful, red = failed) which allowed comparison of landing outcomes at each launch site
- Used mouse position coordinates to identify nearby geographic features which helped capture coordinates of the coastline, city, railway, and highway
- Used blue polylines and distance markers to show proximity to nearby landmarks which enabled analysis of how distance to features like cities and coastlines may impact launch site selection
- External Reference: GitHub Repository – [Launch Site Map with Folium]



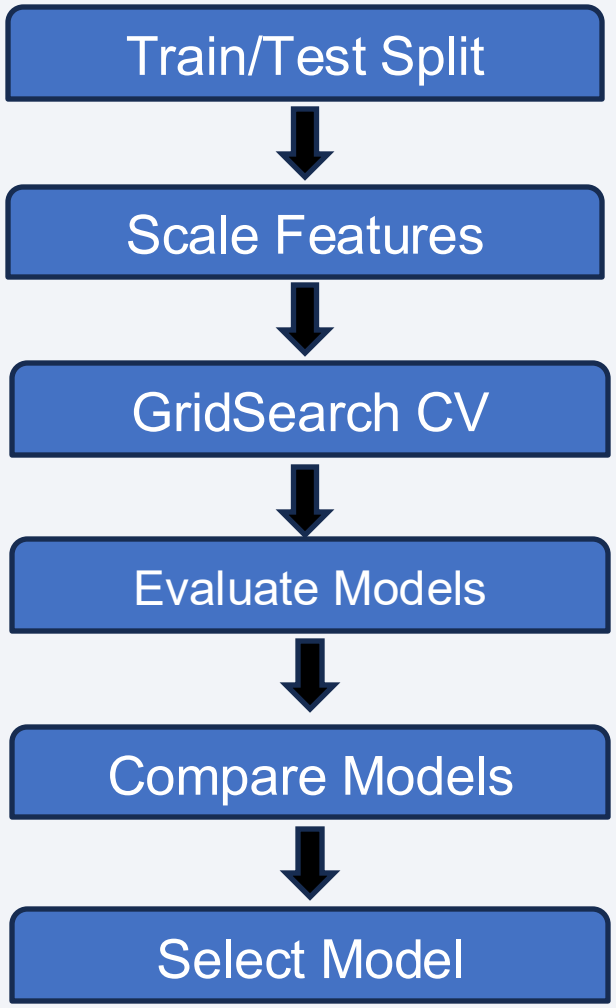
Launch sites are located close to coastlines and transportation infrastructure, while maintaining distance from nearby cities

Build a dashboard with Plotly Dash

- Built an interactive dashboard with a launch site dropdown filter which allows users to view results for all sites or a specific launch site
- Added a pie chart showing the distribution of successful vs failed launches which provides a quick comparison of landing outcomes by site
- Included a payload range slider to enable filtering of data based on payload mass
- Created a scatter plot (payload mass vs. landing outcome) to show the relationship between payload size and launch success
- Added a bar chart (booster version vs. count of landing outcomes) to display number of successful and failed launches by booster version
- Bar chart visual improves comparison across categories with a clearer visualization
- External Reference: GitHub Repository – Plotly Dash Notebook (Scatterplot) and Plotly Dash Notebook (Barchart)

Predictive analysis (classification)

- Split the dataset into training and testing sets
- Applied StandardScaler to normalize features and prepare data for modeling
- Trained classification models including logistic regression, SVM, decision tree, and KNN
- Used GridSearchCV with cross-validation to train and tune classification models with optimized hyperparameters
- Evaluated model performance on the test dataset
- Compared models to identify the best-performing approach
- **Key Finding**: All models achieved similar performance, so logistic regression was selected for it's simplicity and interpretability
- External Reference: GitHub Repository – [Machine Learning Prediction Lab]



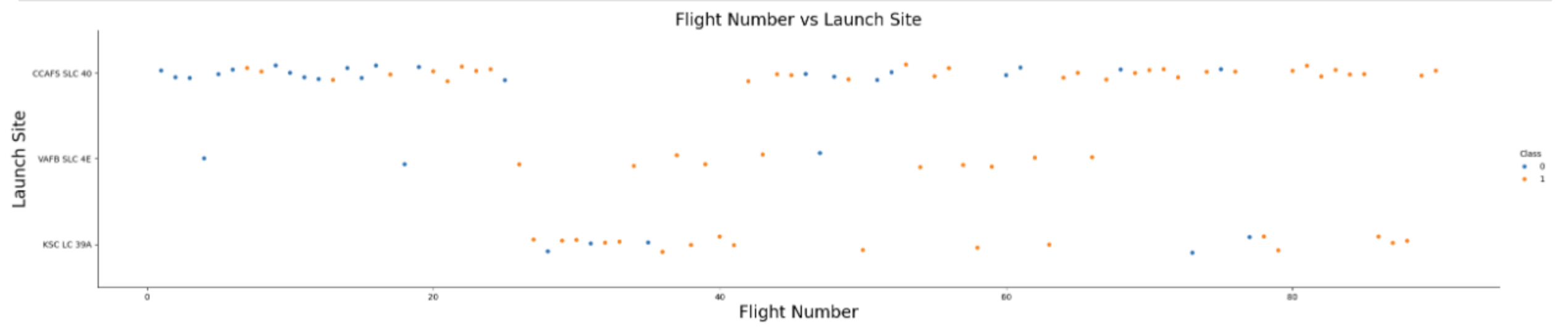
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

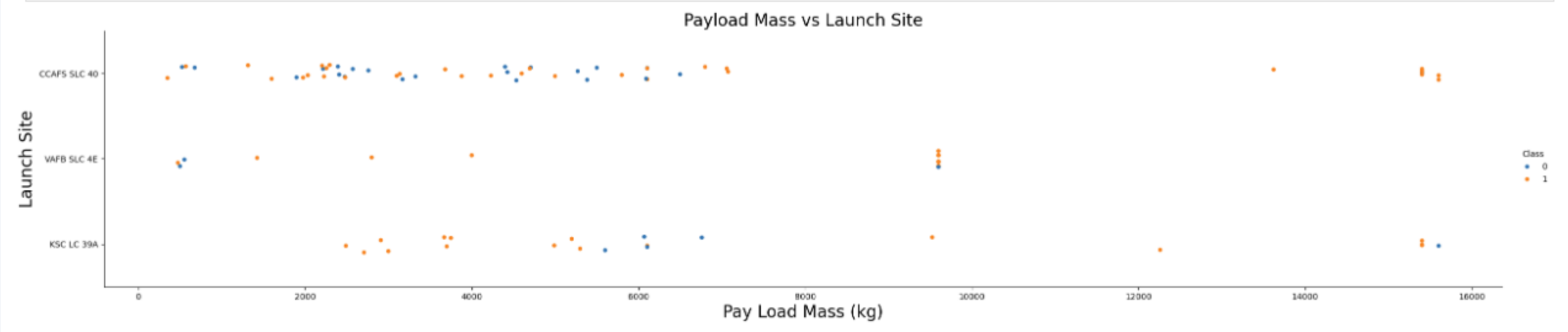
Insights drawn from EDA

Flight number vs. launch site



- This plot shows the relationship between flight number and launch site, with color indicating landing outcome (blue = failure, orange = success).
- CCAFS SLC 40 shows a higher proportion of failed landings, while KSC LC-39A and VAFB SLC-4E display more similar distributions of outcomes when compared to each other.

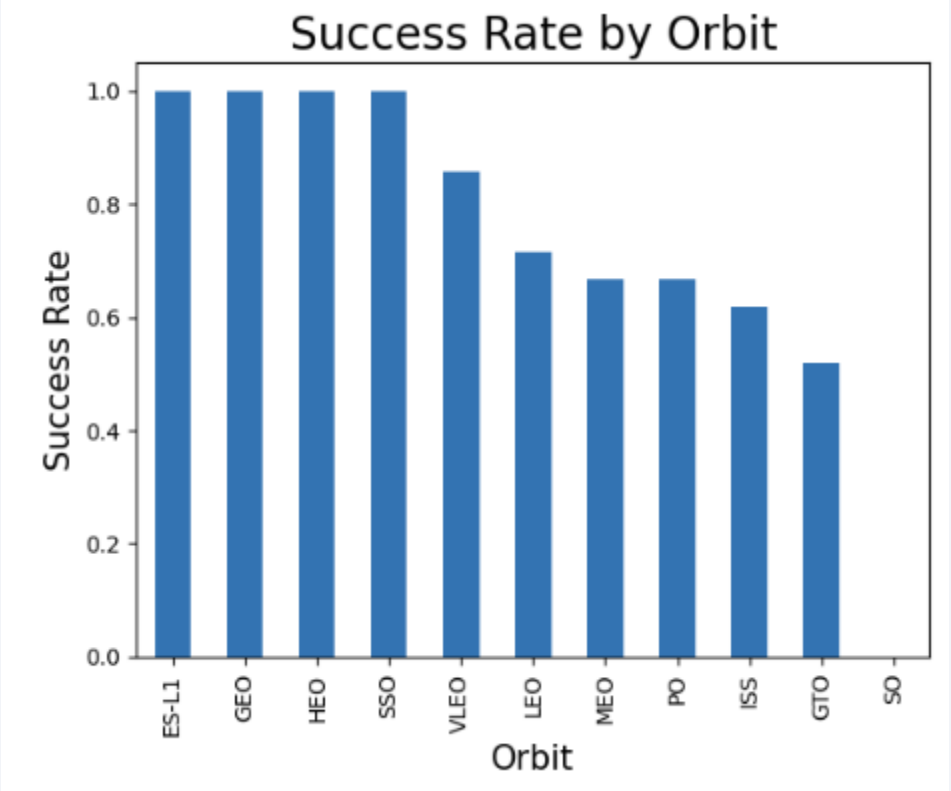
Payload vs. launch site



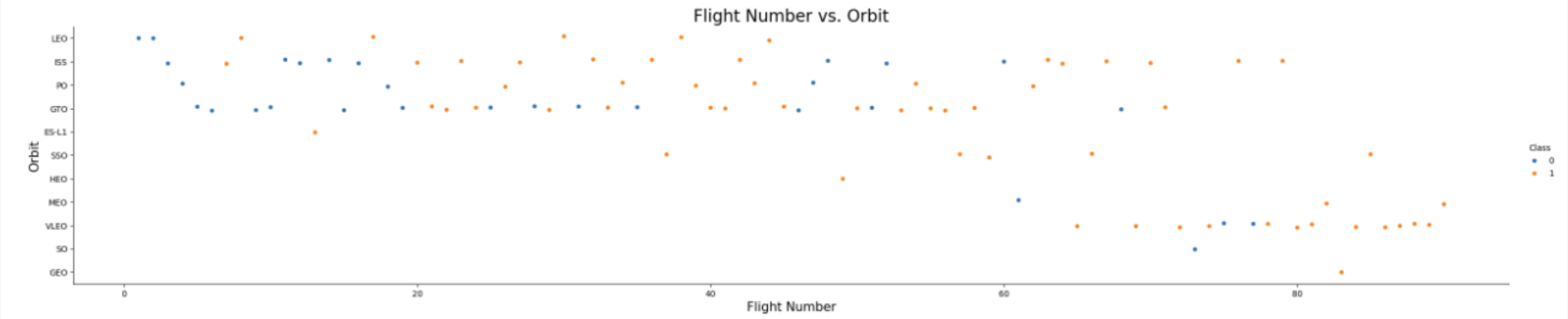
- This plot shows the relationship between pay load mass and launch site, with color indicating landing outcome (blue = failure, orange = success).
- No launches at VAFB-SLC exceed 10,000 kg payload mass, and launches above this threshold generally show high success rates.

Success rate vs. orbit type

- This plot shows the relationship between orbit type and landing success rate
- Landing success varies by orbit type, with ES-L1, GEO, HEO, and SSO showing the highest success rates, suggesting orbit selection may impact landing performance

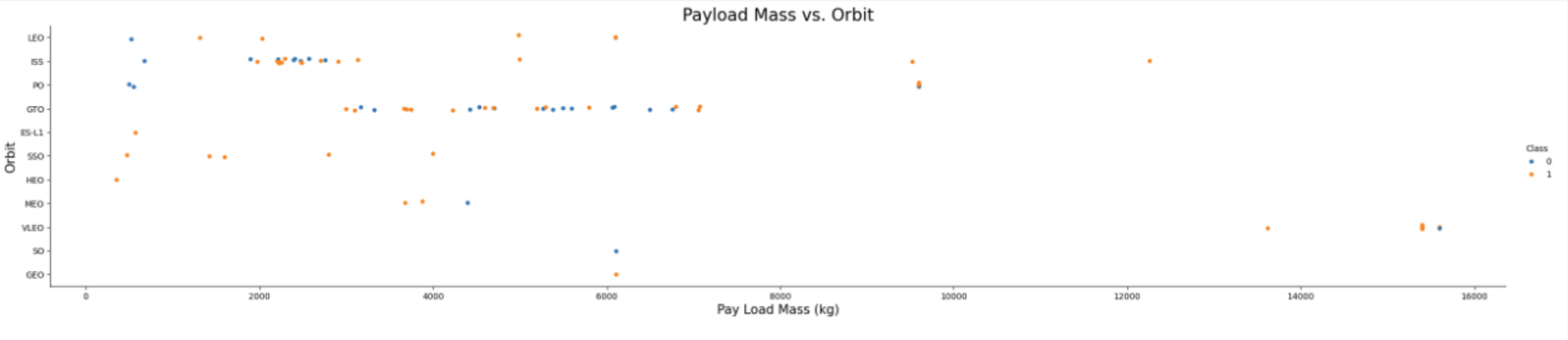


Flight number vs. orbit type



- This plot shows the relationship between flight number and orbit type, with color indicating landing outcome (blue = failure, orange = success).
- Higher flight numbers, representing later missions, show increased use of orbit types such as SO, VLEO, and HEO, with many of these launches achieving successful landings.

Payload vs. orbit type

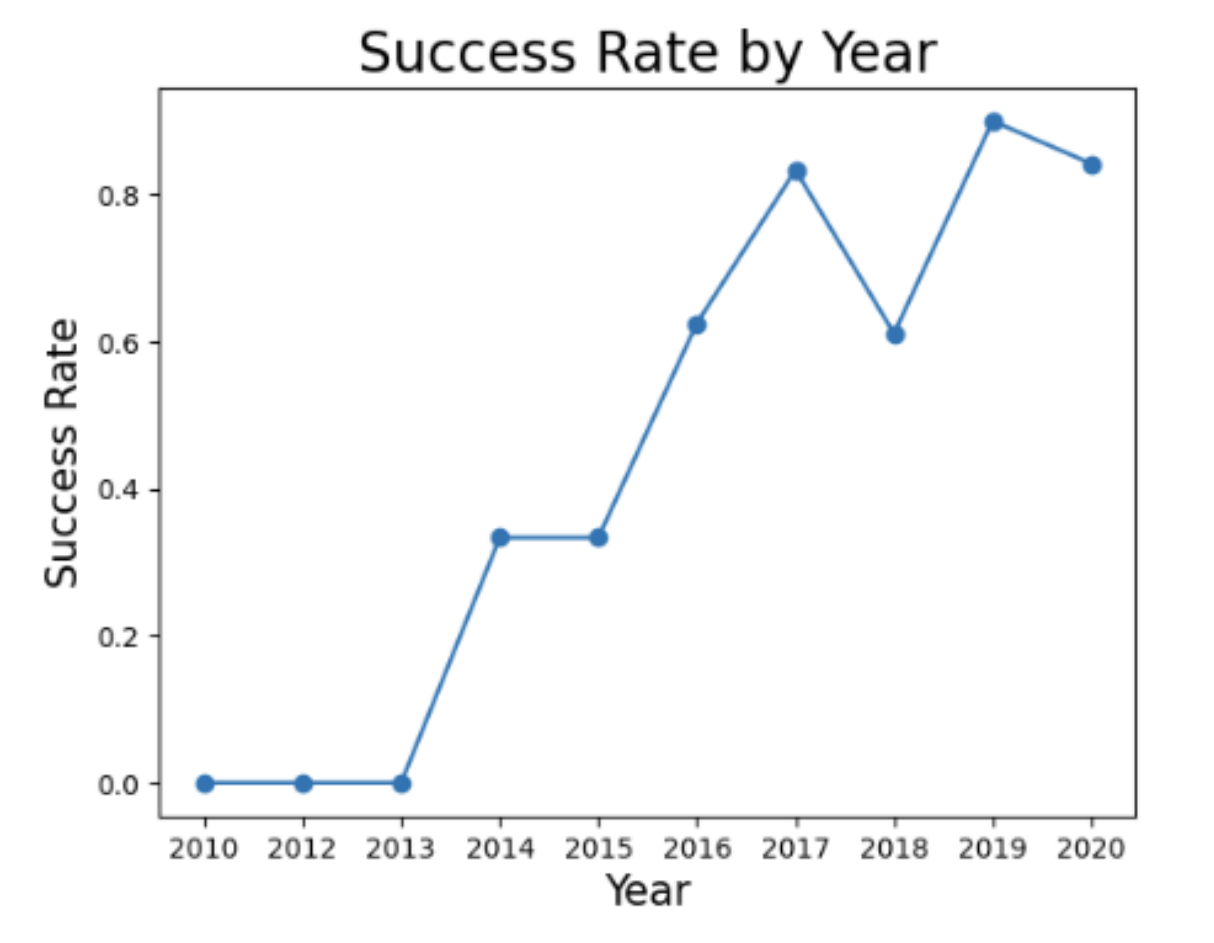


- This plot shows the relationship between pay load mass and orbit type, with color indicating landing outcome (blue = failure, orange = success).
- Orbit types such as LEO, SSO, and MEO are associated with smaller payload masses, while VLEO involves larger payloads. SSO missions show consistently successful outcomes.

Launch success yearly trend

- This plot shows the relationship between the year of launch and success rate

- Success rates generally increase over time, with a dip in 2018, a rise again in 2019, and a slight decrease in 2020.



All launch site names

- Displays the unique launch site names included in the dataset

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch site names begin with “KSC”

- Displays 5 records where the launch site name starts with “KSC,” showing the KSC LC-39A launch site

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

Total payload mass

- Calculates the total payload mass for NASA (CRS) missions, with a result of 45,596 kg

Customer	Total_Payload_Mass
NASA (CRS)	45596

Average payload mass by F9 v1.1

- Calculates the average payload mass for booster version F9 v1.1, with a result of 2,928.4 kg

Booster_Version	Average_Payload_Mass
F9 v1.1	2928.4

First successful ground landing date

- Identifies the date of the first successful drone ship landing, which occurred on 2016-04-08

MIN(Date)	Landing_Outcome
2016-04-08	Success (drone ship)

Successful drone ship landing with payload between 4,000 and 6,000

- Identifies boosters with successful drone ship landings and payload masses within the 4,000–6,000 kg range

Date	Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
2017-05-01	F9 FT B1032.1	5300	Success (ground pad)
2017-09-07	F9 B4 B1040.1	4990	Success (ground pad)
2018-01-08	F9 B4 B1043.1	5000	Success (ground pad)

Total number of successful and failed mission outcomes

- Calculates the total number of mission outcomes, showing 100 successful launches and 1 failure

Mission_Outcome	Number_Mission_Outcomes
Successful	100
Failure	1

Boosters carried maximum payload

- Lists 12 boosters that carried the maximum payload mass of 15,600 kg

Date	Booster_Version	PAYLOAD_MASS__KG_
2019-11-11	F9 B5 B1048.4	15600
2020-01-07	F9 B5 B1049.4	15600
2020-01-29	F9 B5 B1051.3	15600
2020-02-17	F9 B5 B1056.4	15600
2020-03-18	F9 B5 B1048.5	15600
2020-04-22	F9 B5 B1051.4	15600
2020-06-04	F9 B5 B1049.5	15600
2020-09-03	F9 B5 B1060.2	15600
2020-10-06	F9 B5 B1058.3	15600
2020-10-18	F9 B5 B1051.6	15600
2020-10-24	F9 B5 B1060.3	15600
2020-11-25	F9 B5 B1049.7	15600

2017 launch records

- Displays records for 2017 showing month names, successful ground pad landing outcomes, booster versions, and launch sites

Year	MonthName	Landing_Outcome	Booster_Version	Launch_Site
2017	February	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
2017	May	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2017	June	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
2017	August	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
2017	September	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
2017	December	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Rank landing outcomes between 2010-06-04 and 2017-03-20

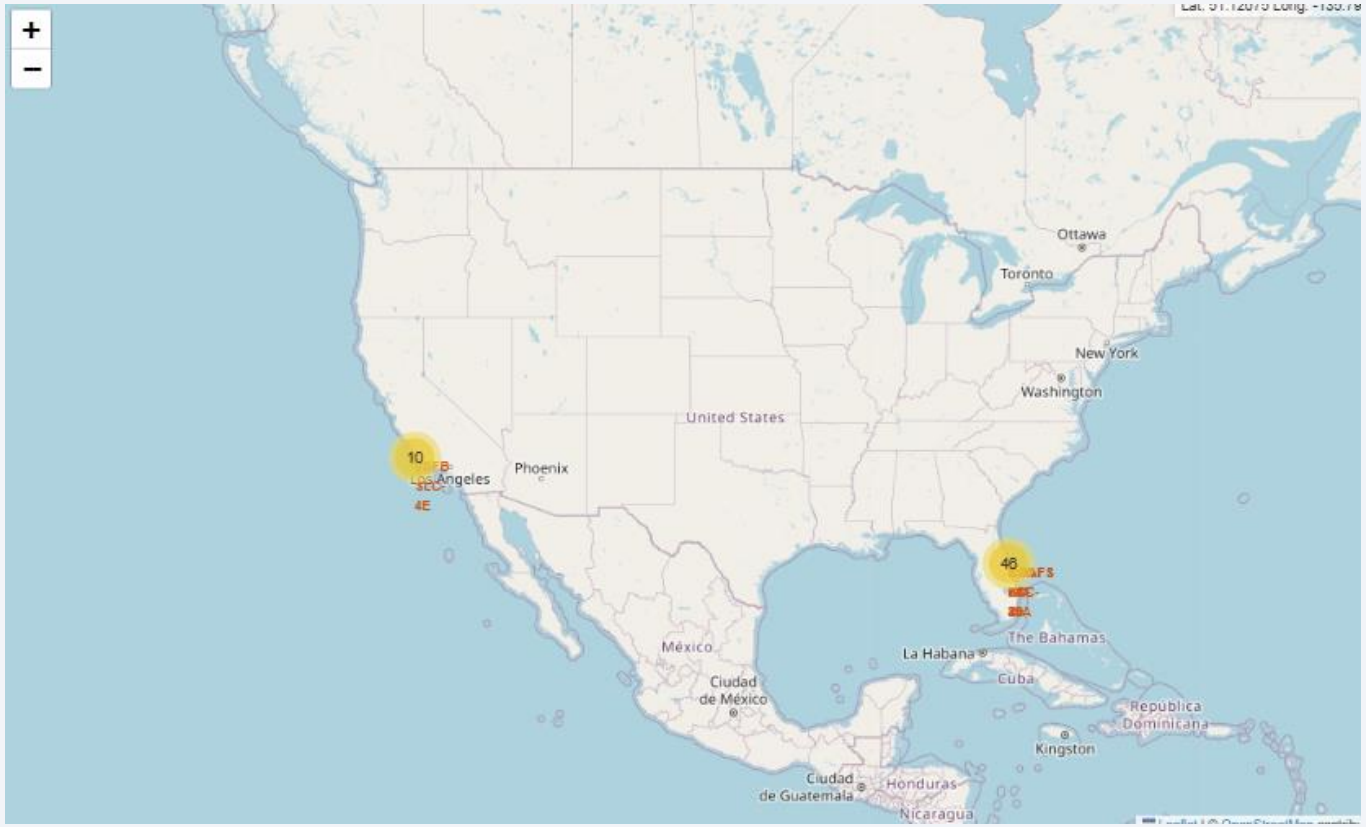
- Ranks landing outcomes by count between 2010-06-04 and 2017-03-20 in descending order

Landing_Outcome	landing_outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

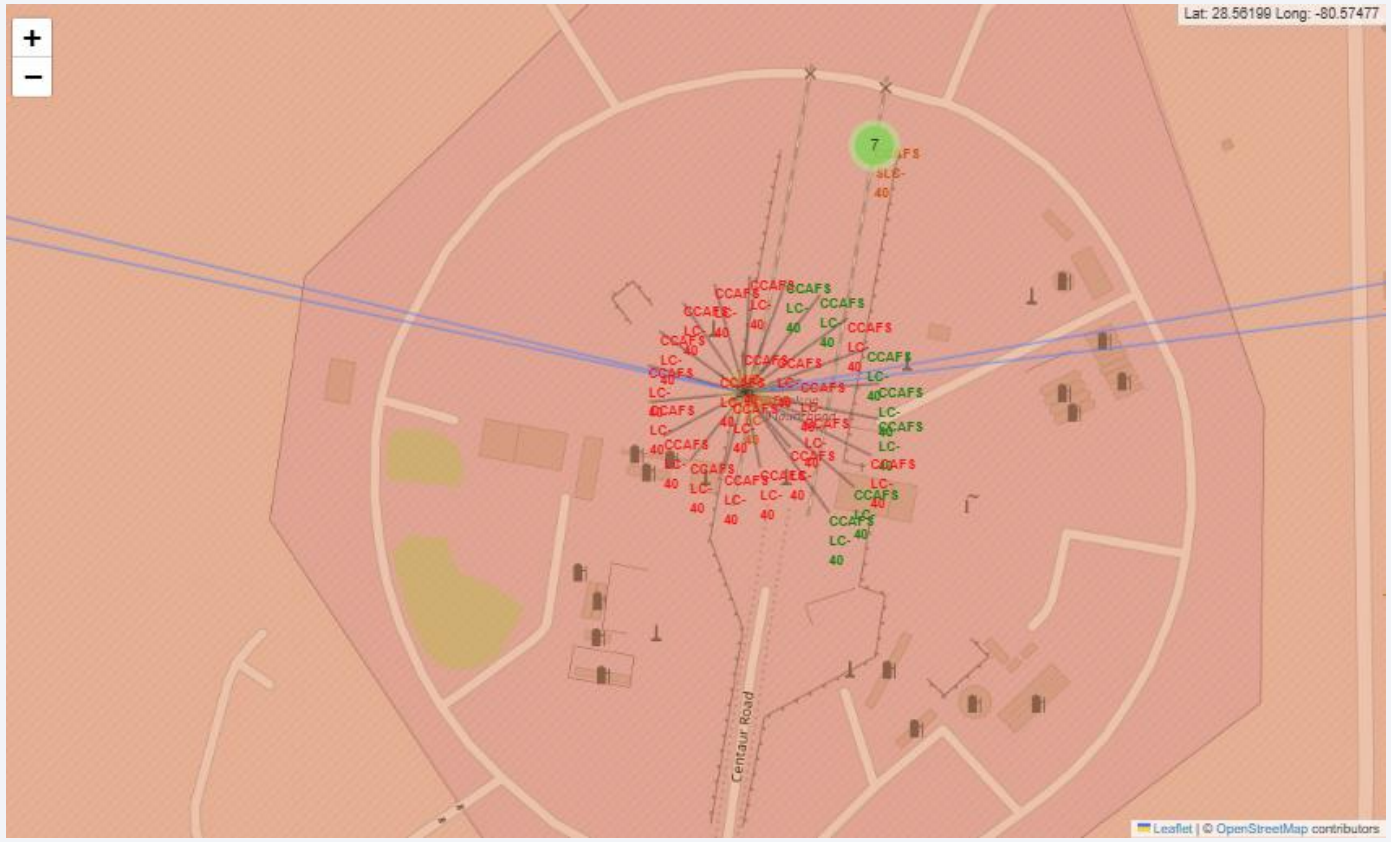
Launch Sites Proximities Analysis

Launch Site Locations



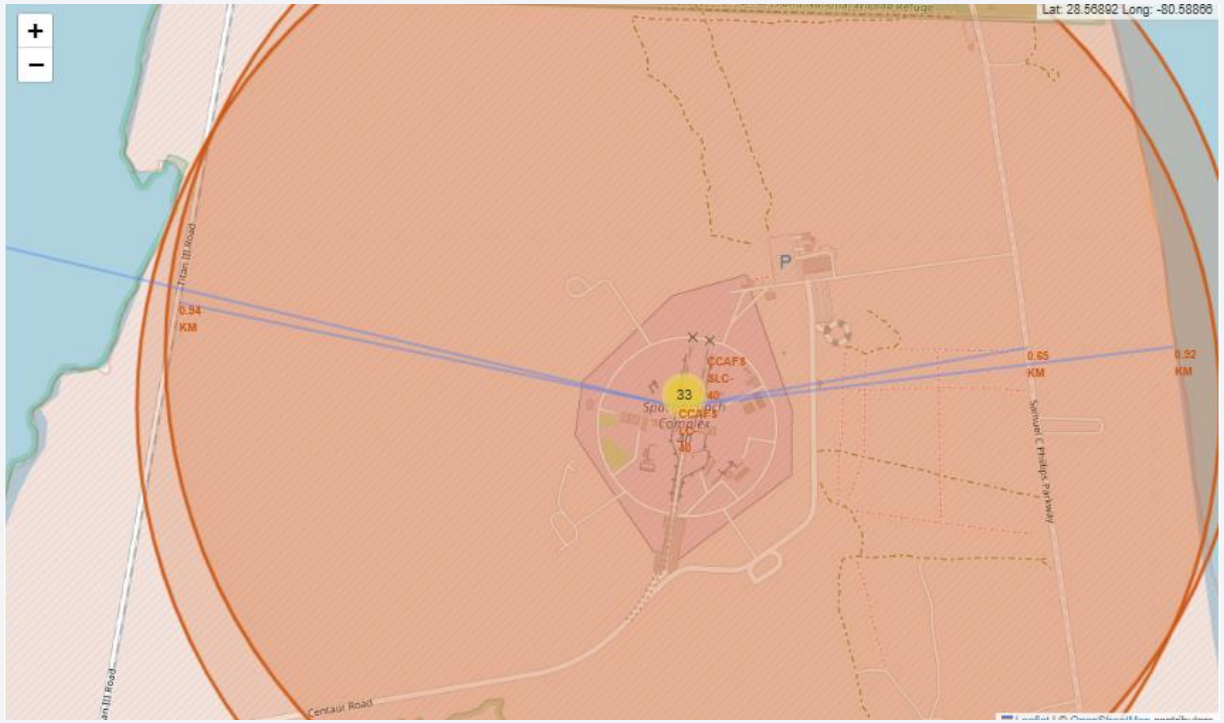
- The map displays four SpaceX launch site locations, with three located within close proximity along the East Coast of the United States and one located on the West Coast.
- External Reference: GitHub Repository – [Space X Folium Notebook]

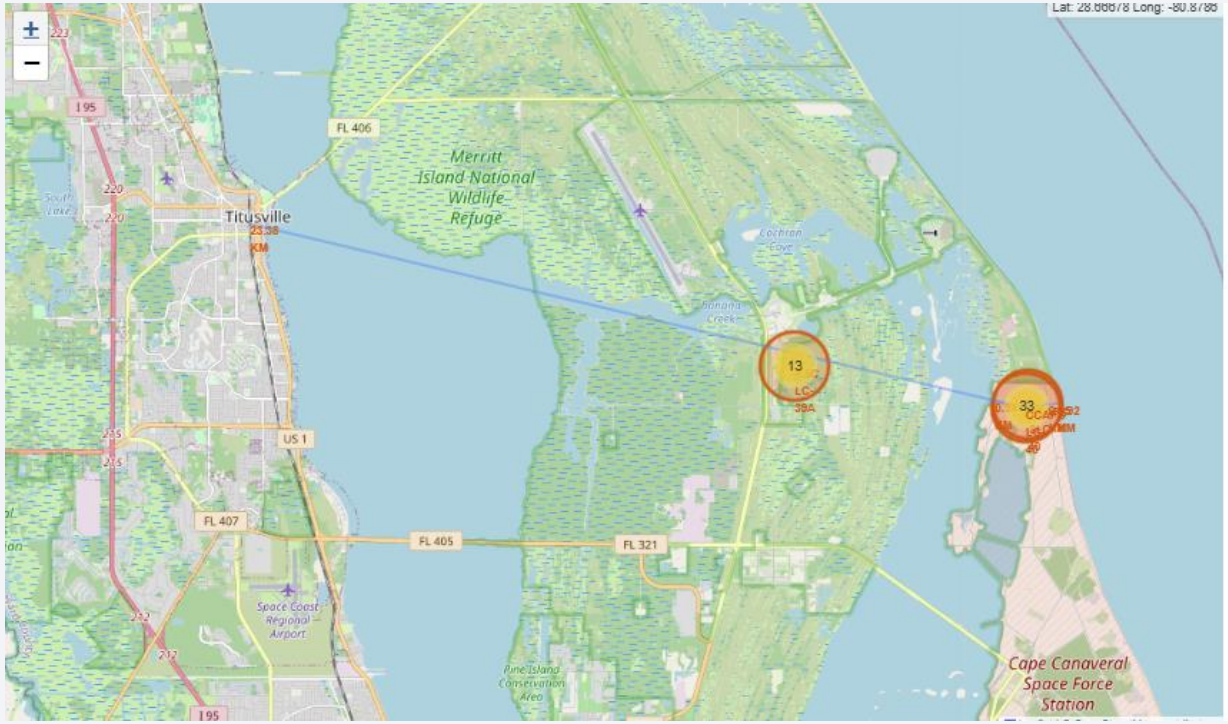
Labeled Launch Outcomes



- The map displays launch outcomes for the CCAFS LC-40 site, showing a higher number of unsuccessful launches (red) compared to successful launches (green).

Launch Site Proximity to Landmarks



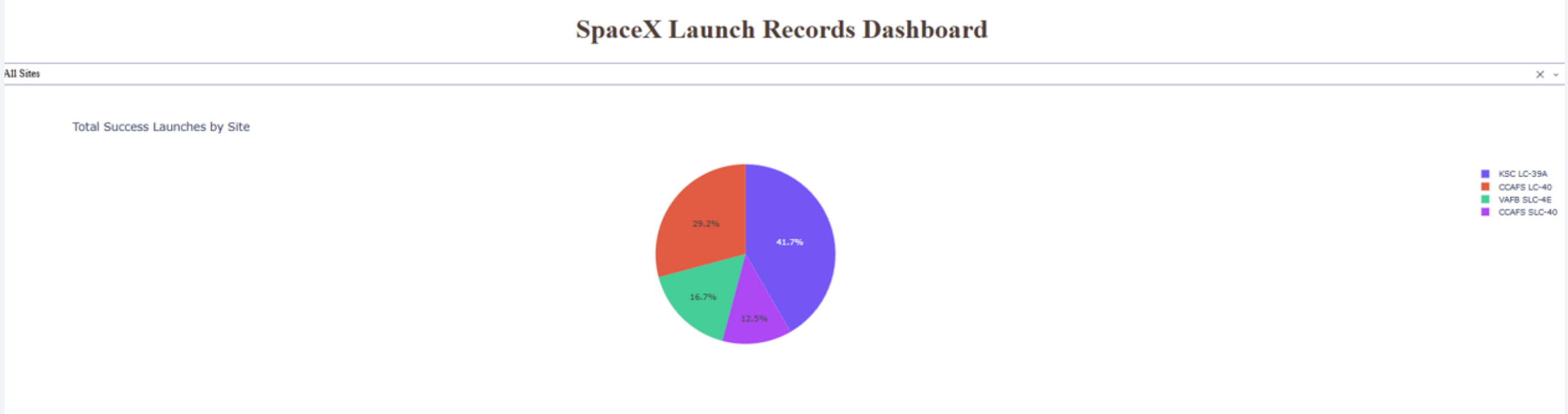


- Launch sites are located close to coastlines and transportation infrastructure, while remaining significantly farther from nearby cities.

Section 4

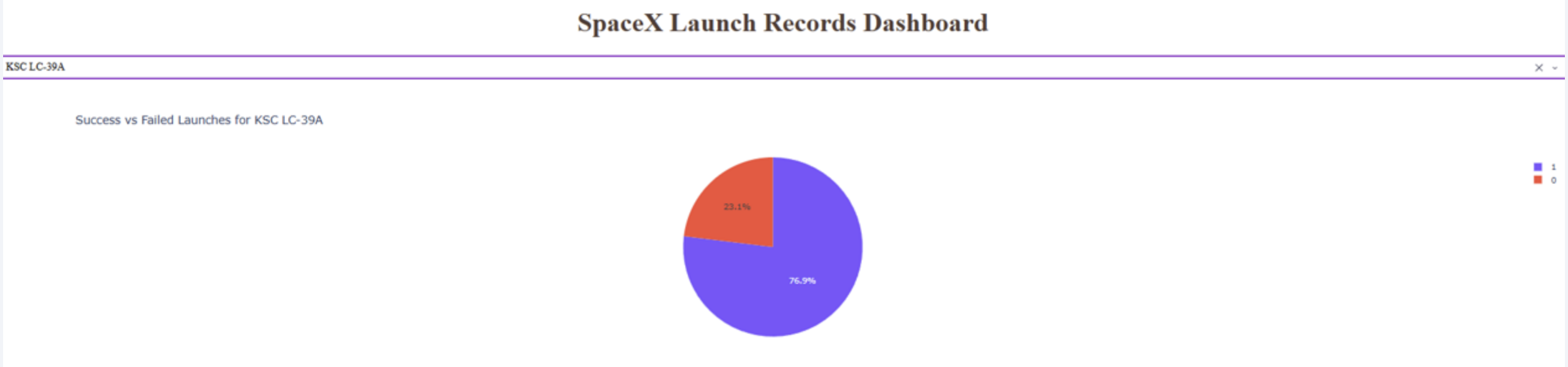
Build a Dashboard with Plotly Dash

Launch Success by Launch Site



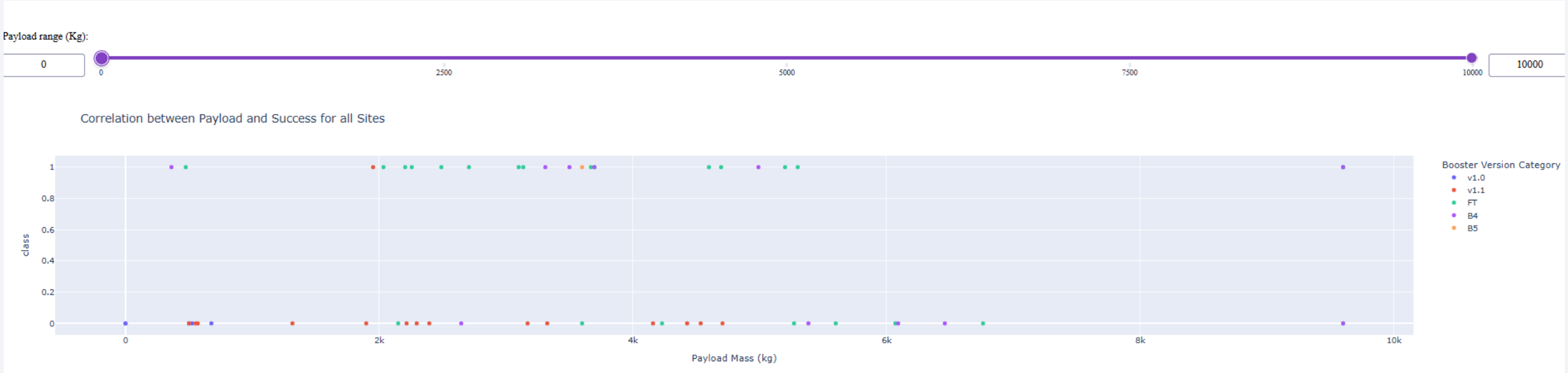
- The pie chart shows success rate for all launch sites with the highest success rate at KSC LC-39A of 41.7% followed CCAFS LC-40, then VAFB SLC-4E, finally CCAFS SLC-40
- External Reference: GitHub Repository – [SpaceX Dash Notebook]

Launch Success for KSC LC-39A



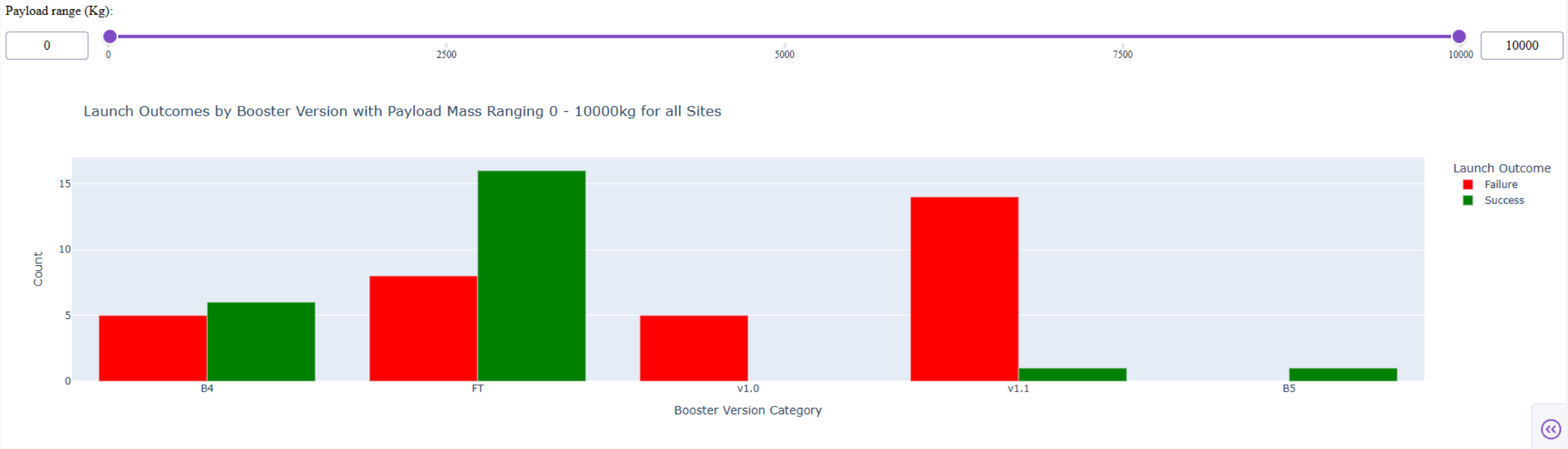
- KSC LC-39A has the highest success rate among all launch sites. Its outcomes consist of a 76.9% success rate and a 23.1% failure rate.

Payload vs Launch Outcome



- Analyzes how payload mass (x-axis) relates to landing success rate (y-axis), with different colors indicating booster versions.

Launch Outcomes by Booster Version



- The bar chart provides a clearer comparison of launch outcomes by booster version compared to the scatter plot. It shows that FT has the highest number of successful launches, while v1.1 has the highest number of failures. B5 exhibits only successful launches (though few), whereas v1.0 shows only unsuccessful launches.
- External Reference: GitHub Repository – [SpaceX Dash Notebook (Bar Chart)]

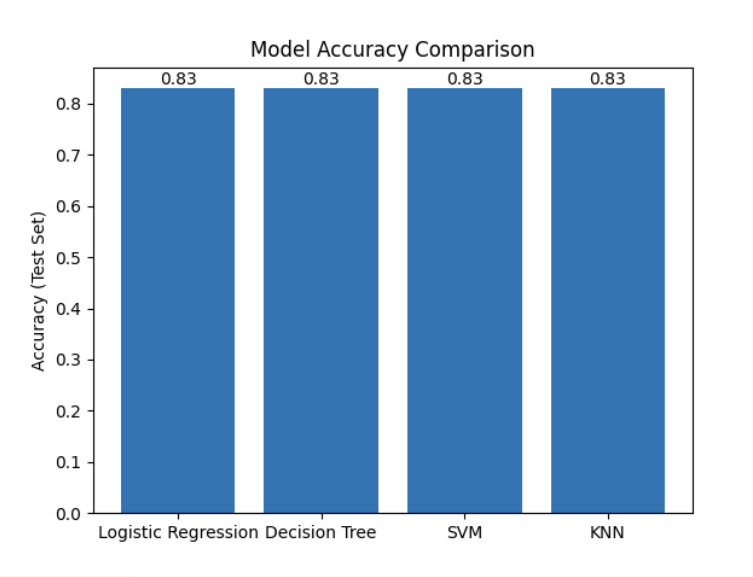
Section 5

Predictive Analysis (Classification)

Classification accuracy

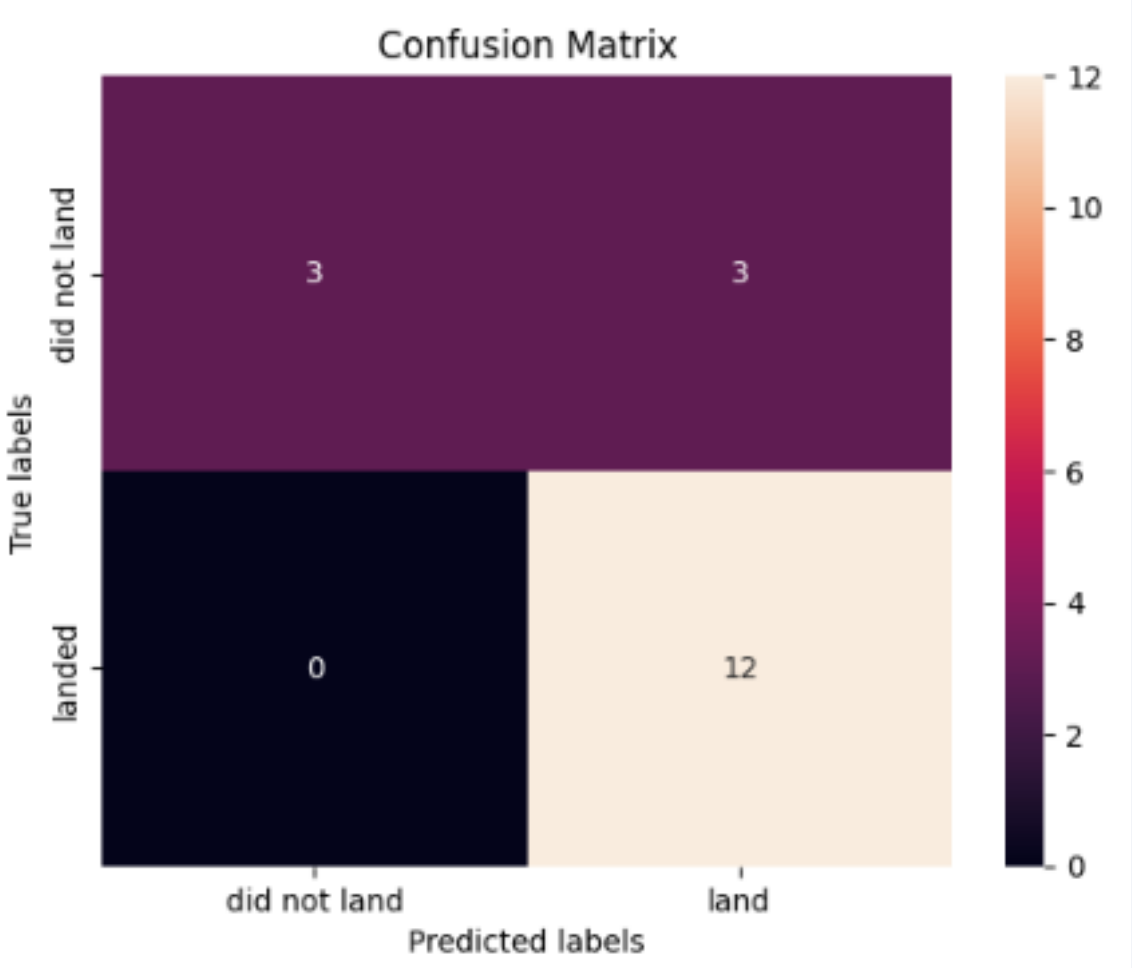
- **Model Evaluation & Selection:** Compared classification accuracy across models including logistic regression, SVM, decision tree, and KNN
- All models achieved similar performance (accuracy ≈ 0.83)
- No single model outperformed the others based on evaluation metrics
- **Model Selection**: Logistic regression was selected as the final model. Chosen for its simplicity, efficiency, and interpretability despite similar performance across models
- External Reference: GitHub Repository – [SpaceX Machine Learning Prediction Notebook]

```
Logistic Regression
              precision    recall  f1-score   support

           0       1.00      0.50      0.67         6
           1       0.80      1.00      0.89        12

    accuracy                           0.83        18
   macro avg       0.90      0.75      0.78        18
weighted avg       0.87      0.83      0.81        18

SVM
              precision    recall  f1-score   support

           0       1.00      0.50      0.67         6
           1       0.80      1.00      0.89        12

    accuracy                           0.83        18
   macro avg       0.90      0.75      0.78        18
weighted avg       0.87      0.83      0.81        18

Decision Tree
              precision    recall  f1-score   support

           0       1.00      0.50      0.67         6
           1       0.80      1.00      0.89        12

    accuracy                           0.83        18
   macro avg       0.90      0.75      0.78        18
weighted avg       0.87      0.83      0.81        18

KNN
              precision    recall  f1-score   support

           0       1.00      0.50      0.67         6
           1       0.80      1.00      0.89        12

    accuracy                           0.83        18
   macro avg       0.90      0.75      0.78        18
weighted avg       0.87      0.83      0.81        18
```



Confusion matrix

- Confusion matrix for the logistic regression model shows 3 misclassifications, all of which are false positives where failed landings were incorrectly predicted as successful.



Conclusions

- Landing success can be predicted with good accuracy (~83%) using machine learning models
- All models performed similarly, so logistic regression was selected for its simplicity and interpretability
- Factors such as payload mass, orbit type, launch site, and flight number influence landing success rates and should be considered in predictive modeling.
- Success rates improve over time, indicating advancements in technology and operations
- Launch sites are strategically located near coastlines and infrastructure while maintaining distance from cities likely for safety reasons

Appendix

- **Code & Notebooks**:
- Full project repository: https://github.com/Kleiner7/SpaceXCapstone/tree/main
- Includes API collection, web scraping, EDA, SQL, visualization, and machine learning notebooks

- **Data Sources:**
- SpaceX API: https://api.spacexdata.com
- Wikipedia (Falcon 9 launches): *https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches*

Thank you!